

CONTINUATION — HelixCode web-IDE platform

Revision: 5 · **Updated:** 2026-07-01 · **Status:** codeserver-1.0.0-dev-0.0.3 RELEASED (real-account SSH-key auth; §11.4.169 matrix 23/23 PASS, 4 mirrors, GH+GL releases). **7 post-release commits** landed on all 4 mirrors toward **codeserver-1.0.0-dev-0.0.4** (login-redirect fix + durability/host-safety hardening + Open VSX marketplace coverage + VS Code Dark default + deep research). Full §11.4.169 matrix release-gate RUNNING NOW → then tag dev-0.0.4. Stack LIVE at <https://192.168.0.213:52443> (all systemd --user).

Read FIRST on any fresh session: this file, then `git fetch --all`, then the auth-pivot spec + AUTH guide + feature ledger below. This is the §12.10 / §11.4.131 standing resumption anchor.

Current phase — post-dev-0.0.3 hardening → codeserver-1.0.0-dev-0.0.4 release-gate RUNNING

codeserver-1.0.0-dev-0.0.3 is **RELEASED**: the real-account SSH-key challenge-response auth model (each session tied to the real host user `milosvasic`, `host-native`), live-validated end-to-end (§11.4.169 matrix **23/23 PASS, 0 FAIL** + Go gate **70 tests -race**), committed (2746e0e), tagged, pushed to all 4 mirrors (no force, §11.4.113), released on GitHub + GitLab.

Since the release, **7 post-release commits** landed on **all 4 mirrors** toward codeserver-1.0.0-dev-0.0.4 (login reachability + durability/host-safety hardening + Open VSX marketplace coverage + default theme + deep research):

- 75e2d9b — login-redirect fix (an unauthenticated browser now lands on the login form instead of a bodyless “This page isn’t working”).
- e0519bf — hardening: **DURABLE** systemd Quadlet edge that survives session crashes + full-host reboot, tarball code-server install (drops the fragile npm path), loopback:8080 host-firewall rule, doc exports.
- 43e37a5 — extensions_auth suite **5/5** (Open VSX plugin install + use, anti-bluff).

- eee6169 — feature ledger reconciled to real evidence (docs/features/Status.md
 - Status_Summary.md).
- 849dc9a — extensions (marketplace) guide + read-only gallery probe.
- 8351e4c — login-form copy/paste clipboard buttons **3/3** + UI-driven marketplace test.
- 97a8c69 — VS Code Dark **default** theme (+ pixel proof) + popular-extensions install/use/persist + auth-modernization deep-research report.

The stack is LIVE + reachable at **https://192.168.0.213:52443** — every component is systemd --user-managed (Caddy edge Quadlet, helix-auth, helix-code-server). All standing operator requests have landed: extension install/use tests, login-form copy/paste buttons, popular-extensions install/use/persist, VS Code Dark default, and the first deep-research report.

- Changelog (in progress): docs/changelogs/codeserver-1.0.0-dev-0.0.4.md
- Prior release: docs/changelogs/codeserver-1.0.0-dev-0.0.3.md
- Auth pivot spec (authoritative): docs/superpowers/specs/2026-07-01-auth-pivot-ssh-key.md
- Extensions guide: docs/guides/EXTENSIONS.md · Edge boot: docs/guides/EDGE_BOOT.md
- User guide: docs/guides/AUTH.md · Feature ledger: docs/features/Status.md
- Deep-research (standing directive): docs/research/auth_modernization_20260701/FINDINGS.md (first report → **WebAuthn/passkeys recommended**)

Why the pivot (captured facts, §11.4.6)

This host is ALT Linux with the **tcb** shadow scheme, so a **non-root** service cannot verify passwords via PAM (/etc/tcb/<user>/shadow unreadable by the user; tcb_chkpwd setuid helper permission-denied to non-root ⇒ pam_authenticate returns PAM_AUTH_ERR for every password). A password gate would need root; the operator directive is **no sudo/root** and **all access via SSH keys**. ~/.ssh/authorized_keys is readable and ssh-keygen -Y sign/-Y verify are supported, so SSH-key challenge-response runs fully **as non-root milosvasic**, with **no password and nothing stored**.

Architecture (this release)

```

Browser —HTTPS→ Caddy (TLS edge, CONTAINERIZED rootless Podman;
HTTP/3 + brotli via custom xcaddy)
                    |
                    |— forward_auth → helix-auth (host-native
Gin, systemd --user, loopback:8081, NON-root)
                    |
                    |— reverse_proxy → code-server (host-native

```

```
AS milosvasic, systemd --user,  
                                --auth none,  
loopback:8080; inherits ~/.ssh, ~/.bashrc, host binaries)
```

- **code-server** — host-native systemd --user, --auth none, loopback:8080.
- **helix-auth** — host-native **Gin** gate, systemd --user, loopback:8081, non-root: SSH-key challenge (GET /login) → verify pasted signature vs authorized_keys (ssh-keygen -Y verify) → __Host-session cookie; GET /auth for forward_auth; **fails closed**.
- **Caddy** — sole containerized component: TLS edge, forward_auth (fail-closed), HTTP/3, encode zstd br gzip, /proxy 403, X-Helix-User strip.
- **Stack directive adopted:** Go + Gin + HTTP/3 (QUIC) + Brotli.

Feature status (§11.4.6 — honest)

The real-account + SSH-key rows in docs/features/Status.md are **PASS with cited captured evidence** (§11.4.169 matrix 23/23 PASS; Status.md Revision 2). Testing: Go unit/integration/race (**70 tests, -race, 81.8% cover**) + shell suites tests/types/{e2e,security,stress_chaos,concurrency,load,memory,benchmark,challenges,helixqa}_auth.sh + banks tests/banks/helixcode-auth-{challenges,helixqa}.yaml. The live aggregate is the conductor-filled placeholder in the changelog.

Honest boundaries carried into the release

- Editor “jail” is **cosmetic** (Explorer default only; terminal/Open-Folder/ extensions keep full host access by design).
- --auth none on loopback:8080 is a **residual** — host firewall rule for loopback:8080 recommended (all real auth is at the gate/edge).
- code-server pinned **4.117.0** (newest on npm at release time).
- Real public Let’s Encrypt remains **operator-gated** (see docs/guides/TLS.md).

Migration

scripts/install-auth.sh (host-native, **no sudo**) provisions the systemd --user code-server + helix-auth units; deploy/up.sh brings up the Caddy edge. **CODE_SERVER_PASSWORD is retired** → HELIX_AUTH_MODE=sshkey, HELIX_AUTH_ACCOUNT, HELIX_AUTH_AUTHORIZED_KEYS, HELIX_AUTH_PRINCIPAL, PROJECTS_ROOT (no password parameter).

Immediate NEXT (to close codeserver-1.0.0-dev-0.0.4)

1. Full §11.4.169 test-type matrix release-gate — **RUNNING NOW** against the live host stack; capture evidence under docs/qa/codeserver-1.0.0-dev-0.0.4/ (§11.4.6 — real PASS/FAIL/SKIP, never invented).
2. Fill the live-validation aggregate in docs/changelogs/codeserver-1.0.0-dev-0.0.4.md from the gate's real counts.
3. Tag codeserver-1.0.0-dev-0.0.4 (§11.4.151 prefix) → push to all 4 mirrors via merge-onto-latest-main (no force-push, §11.4.113) → publish GitHub + GitLab releases.
4. Standing operator directive — **frequent deep research** (§11.4.150): keep producing reports; the first landed report recommends **WebAuthn/passkeys** as the auth-modernization path (docs/research/auth_modernization_20260701/FINDINGS.md).

Binding constraints (every phase)

Port band **52000-52999**; publish only edge ports on 0.0.0.0; **no secrets** (only git-ignored .env); **every gate paired with a §1.1 mutation**; **no --force/--no-verify/bypass**; hardlinked .git backup before destructive ops; **anti-bluff** (PASS needs captured runtime evidence, no invented numbers); stop + root-cause + full retest on any defect; release prefix codeserver; submodule commits propagate first; docs kept in sync.

Prior state (Phases 1-2, retained)

Phase 1 COMPLETE; Phase 2 core working+verified (Caddy edge + code-server stack on Podman; TLS1.3 edge on 52443, →301 on 52080; \$PROJECTS bind-mounted). Owned submodules (8) at root carry the Constitution inheritance pointer; constitution pinned (see history). Prior release codeserver-1.0.0-dev-0.0.2 shipped Let's Encrypt HTTPS (auto-renew + rotation) + the 14-suite §11.4.169 matrix.